1.1.41

Signal Processing – Analog

EE 2007

Question

If $u(t)$, $r(t)$ denote the unit step and unit ramp functions respectively and $u(t) * r(t)$ their convolution, then the function $u(t+1) * r(t-2)$ is given by

a) $\frac{1}{2}(t-1)(t-2)$

b) $\frac{1}{2}(t-1)(t-2)$

c) $\frac{1}{2}(t-1)^2\, u(t-1)$

d) None of the above

General Setup: Shifted Signals

Define general time-shifted unit step and unit ramp:

$$u(t-a) = \begin{cases} 1, & t \geq a \\ 0, & t < a \end{cases} \tag{1}$$

$$r(t-b) = (t-b)\ u(t-b) = \begin{cases} t-b, & t \geq b \\ 0, & t < b \end{cases} \tag{2}$$

We compute the general convolution:

$$y(t) = u(t-a) * r(t-b) = \int_{-\infty}^{\infty} u(\tau - a)\, r(t-\tau-b)\ d\tau \tag{3}$$

Goal

Derive a closed-form for any $a, b \in \mathbb{R}$, then substitute $a = -1,\ b = 2$.

General Solution: Limits of Integration

Both factors must be nonzero simultaneously:

$$u(\tau - a) \neq 0 \quad \Rightarrow \quad \tau \geq a \tag{4}$$

$$r(t - \tau - b) \neq 0 \quad \Rightarrow \quad t - \tau - b \geq 0 \quad \Rightarrow \quad \tau \leq t - b \tag{5}$$

Active integration range: $a \leq \tau \leq t - b$, which requires:

$$a \leq t - b \quad \Rightarrow \quad t \geq a + b \tag{6}$$

$y(t) = 0$ for $t < a + b$. Output starts at $t = a + b$.

General Solution: Evaluating the Integral

For $t \geq a + b$, with $u(\tau - a) = 1$ over $[a,\ t - b]$:

$$y(t) = \int_a^{t-b} (t - \tau - b)\ d\tau \tag{7}$$

Let $s = t - \tau - b$, so $ds = -d\tau$:

- When $\tau = a$: $\quad s = t - a - b$
- When $\tau = t - b$: $\quad s = 0$

$$y(t) = \int_{t-a-b}^{0} s \cdot (-ds) = \int_0^{t-a-b} s\ ds \tag{8}$$

$$y(t) = \left[\frac{s^2}{2}\right]_0^{t-a-b} = \frac{(t - a - b)^2}{2} \tag{9}$$

General Result

Combining both cases, the **general convolution formula** is:

$$\boxed{u(t-a) * r(t-b) = \frac{1}{2}(t-a-b)^2\, u(t-a-b)} \tag{10}$$

Key observations:

- The output is always a **quadratic ramp** — same shape as $u(t) * r(t)$
- It is simply the base case $\frac{1}{2}t^2 u(t)$ **shifted right by** $a+b$
- $a < 0$ (step advanced) **reduces** the onset time
- $b > 0$ (ramp delayed) **increases** the onset time
- Net shift $= a + b$

Applying to the Question

Substitute $a = -1$ (since $u(t+1) = u(t-(-1))$) and $b = 2$:

$$u(t+1) * r(t-2) = \frac{1}{2}(t-(-1)-2)^2 \; u(t-(-1)-2) \tag{11}$$

$$= \frac{1}{2}(t-1)^2 \; u(t-1) \tag{12}$$

Net shift $= a + b = -1 + 2 = +1$ (output starts at $t = 1$)

Answer

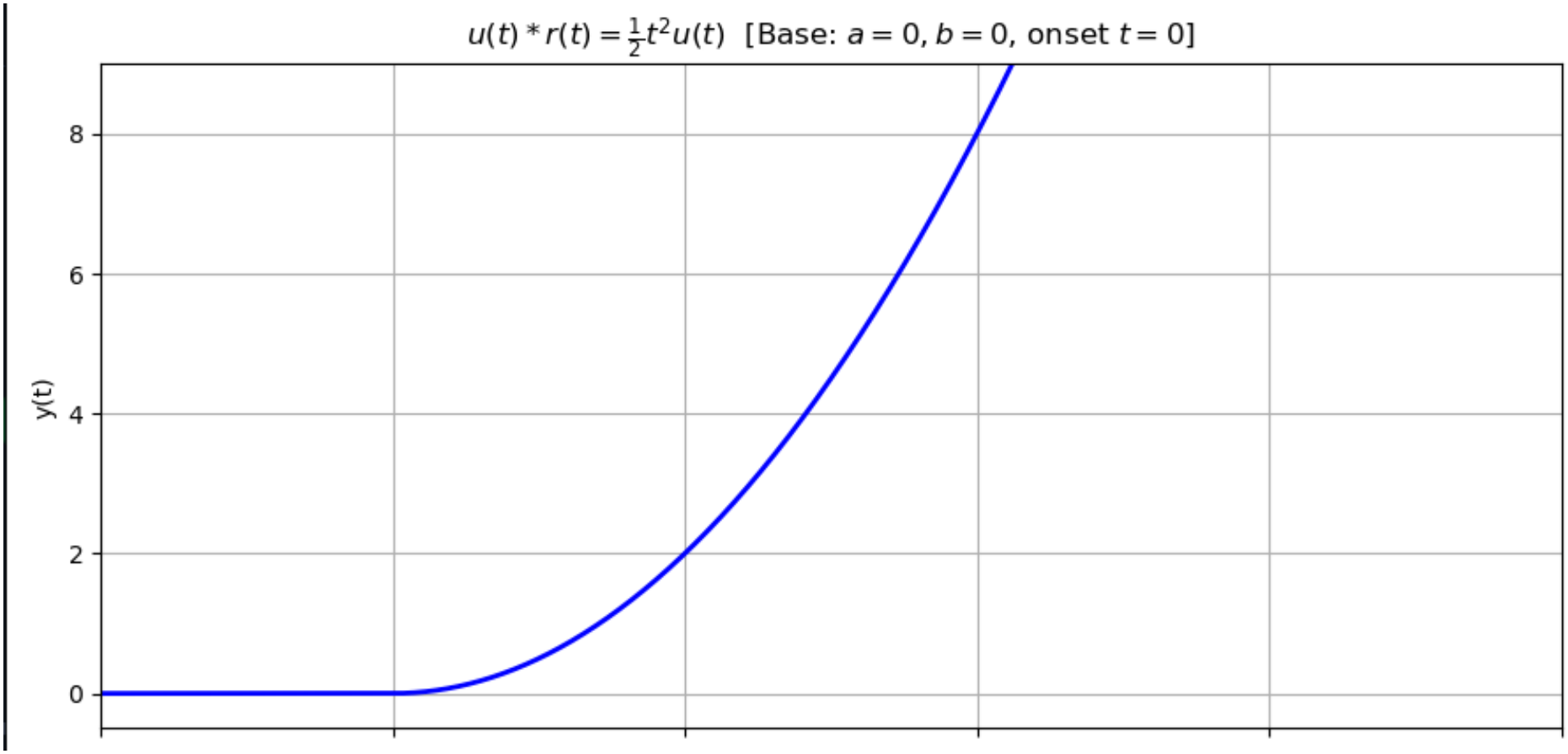
Option (c): $\frac{1}{2}(t-1)^2 \, u(t-1)$

Relation Between the Three Cases

Case	a	b	**Result**
Base: $u(t) * r(t)$	0	0	$\frac{1}{2}t^2\, u(t)$, onset $t = 0$
General: $u(t-a) * r(t-b)$	a	b	$\frac{1}{2}(t-a-b)^2\, u(t-a-b)$
Question: $u(t+1) * r(t-2)$	-1	2	$\frac{1}{2}(t-1)^2\, u(t-1)$, onset $t = 1$

All three are **identical parabolic shapes** – only the starting point on the t-axis differs by $a + b$. The question's curve is the base case $u(t) * r(t)$ shifted **right by 1 unit**.

Plot



Plot

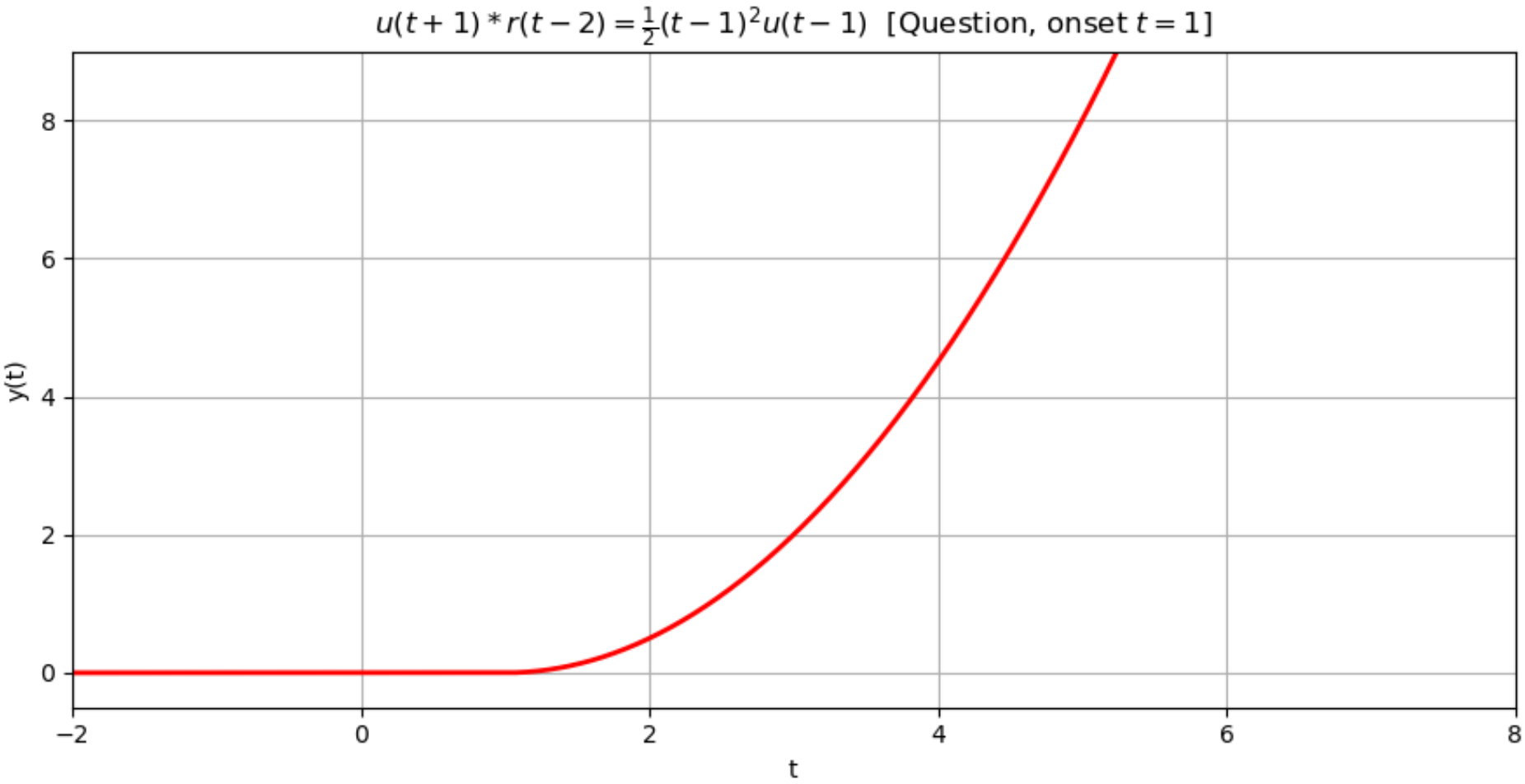


Figure: Blue: base $u(t) * r(t)$, Green: general $(a = -1, b = 2)$, Red: question $u(t+1) * r(t-2)$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

def u(t):
    return np.where(t >= 0, 1.0, 0.0)

t = np.linspace(-2, 8, 10000)

y_base = 0.5 * t**2 * u(t)

a, b = -1, 2
y_general = 0.5 * (t - a - b)**2 * u(t - a - b)

y_question = 0.5 * (t - 1)**2 * u(t - 1)

fig, axes = plt.subplots(2, 1, figsize=(9, 9), sharex=True)
```

Python Code

```
axes[0].plot(t, y_base, 'b-', linewidth=2)
axes[0].set_title(r'$u(t)*r(t)=\frac{1}{2}t^2 u(t)$ [Base: $a=0,b
    =0$, onset $t=0$]')
axes[0].set_ylabel('y(t)'); axes[0].grid(True); axes[0].set_ylim
    (-0.5,9)

axes[1].plot(t, y_question, 'r-', linewidth=2)
axes[1].set_title(r'$u(t+1)*r(t-2)=\frac{1}{2}(t-1)^2 u(t-1)$ [
    Question, onset $t=1$]')
axes[1].set_ylabel('y(t)'); axes[1].set_xlabel('t')
axes[1].grid(True); axes[1].set_ylim(-0.5,9)

plt.xlim(-2, 8)
plt.tight_layout()
plt.savefig("image.png", dpi=300)
plt.show()
```